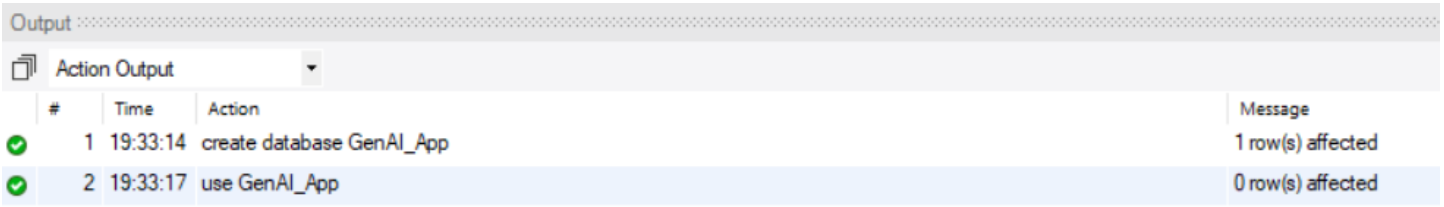
GenAI Application – SQL Data Types and Constraints

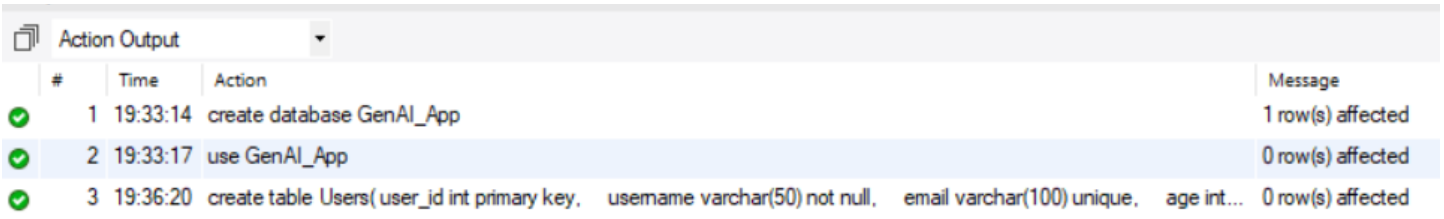
1. **Create Database: GenAI_App**

```
create database GenAI_App;
use GenAI_App;
```



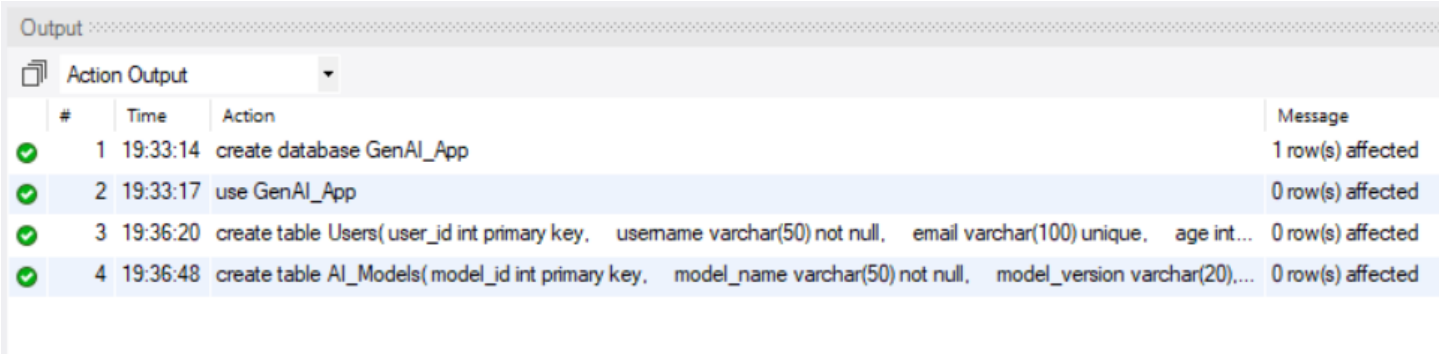
2. **Create a table named Users with the following columns:**
   - **user_id – Integer, Primary Key**
   - **username – Maximum 50 characters, cannot be NULL**
   - **email – Maximum 100 characters, must be unique**
   - **age – Integer, must be 18 or above**
   - **phone – Fixed 10 characters**
   - **registration_date – Date**
   - **account_status – Default value should be 'Active'**

```
create table Users(
        user_id int primary key,
        username varchar(50) not null,
        email varchar(100) unique,
        age int check (age >= 18),
        phone varchar(10),
        registration_date date,
        account_status varchar(50) default 'Active'
);
```



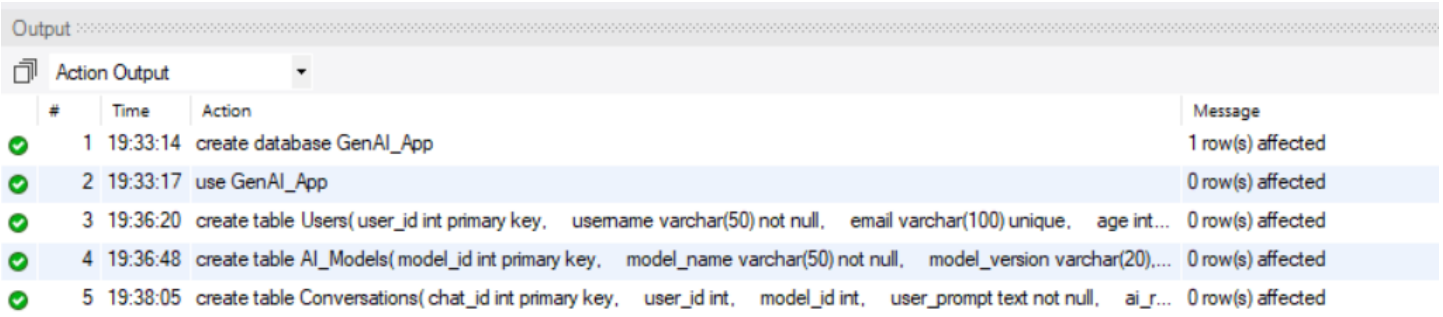
3. **Create a table named AI_Models with:**
   - **model_id – Integer, Primary Key**
   - **model_name – Maximum 50 characters, cannot be NULL**
   - **model_version – Maximum 20 characters**
   - **is_available – Boolean, default value TRUE**

```
create table AI_Models(
        model_id int primary key,
        model_name varchar(50) not null,
        model_version varchar(20),
        is_available boolean default TRUE
);
```



4. **Create a table named Conversations with:**
   - **chat_id – Integer, Primary Key**
   - **user_id – Integer, Foreign Key referencing Users**
   - **model_id – Integer, Foreign Key referencing AI_Models**
   - **user_prompt – Text, cannot be NULL**
   - **ai_response – Text**
   - **token_count – Integer, must be zero or above**
   - **chat_time – Date and Time**

```
create table Conversations(
        chat_id int primary key,
        user_id int,
        model_id int,
        user_prompt text not null,
        ai_response text,
        token_count int check (token_count >= 0),
        chat_time datetime,
        foreign key (user_id) references Users(user_id),
        foreign key (model_id) references AI_Models(model_id)
);
```



5. **Payments Table**

```
create table Payments (
    payment_id int primary key,
    user_id int,
    amount decimal(10,2),
    payment_date datetime,
    payment_status varchar(20) default 'Pending',
    foreign key (user_id) references Users(user_id)
    );
```

Output

Action Output

#	Time	Action	Message
1	19:33:14	create database GenAI_App	1 row(s) affected
2	19:33:17	use GenAI_App	0 row(s) affected
3	19:36:20	create table Users(user_id int primary key, username varchar(50) not null, email varchar(100) unique, age int...	0 row(s) affected
4	19:36:48	create table AI_Models(model_id int primary key, model_name varchar(50) not null, model_version varchar(20),...	0 row(s) affected
5	19:38:05	create table Conversations(chat_id int primary key, user_id int, model_id int, user_prompt text not null, ai_r...	0 row(s) affected
6	19:41:55	create table Payments (payment_id int primary key, user_id int, amount decimal(10,2), payment_date date...	0 row(s) affected

Queries and Tasks

1. Display all records from the four tables

select * from Users;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap

user_id	username	email	age	phone	registration_date	account_status
1	Arun	arun@gmail.com	22	9876543210	2026-01-10	Active
2	Bala	bala@gmail.com	28	9876543211	2026-01-12	Active
3	Charan	charan@gmail.com	19	9876543212	2026-02-05	Inactive
4	David	david@gmail.com	32	9876543213	2026-02-15	Active
5	Ezhil	ezhil@gmail.com	25	9876543214	2026-03-01	Active
6	Farhan	farhan@gmail.com	30	9876543215	2026-03-10	Inactive
7	Gokul	gokul@gmail.com	21	9876543216	2026-04-02	Active
8	Hari	hari@gmail.com	27	9876543217	2026-04-12	Active
9	Imran	imran@gmail.com	35	9876543218	2026-05-01	Active
10	John	john@gmail.com	24	9876543219	2026-05-15	Inactive
NULL	NULL	NULL	NULL	NULL	NULL	NULL

select * from AI_Models;

Result Grid | Filter Rows: | Edit:

model_id	model_name	model_version	is_available
101	GPT-4	4.0	1
102	GPT-3.5	3.5	1
103	Gemini	1.5	1
104	Claude	3.0	1
105	Llama	3.1	0
106	Mistral	7B	1
107	DeepSeek	R1	0
108	PaLM	2.0	0
109	Cohere	Command-R	1
110	Phi	3.0	1
NULL	NULL	NULL	NULL

select * from Conversations;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

chat_id	user_id	model_id	user_prompt	ai_response	token_count	chat_time
1001	1	101	Explain Python	Python is a programming language.	250	2026-06-01 09:30:00
1002	2	102	Explain SQL	SQL is used to manage databases.	350	2026-06-02 10:15:00
1003	3	103	What is AI?	AI stands for Artificial Intelligence.	600	2026-06-03 11:00:00
1004	4	104	Explain Machine Learning	ML enables computers to learn from data.	750	2026-06-04 12:30:00
1005	5	105	What is Deep Learning?	Deep Learning uses neural networks.	450	2026-06-05 13:45:00
1006	6	106	Explain RAG	RAG combines retrieval and generation.	900	2026-06-06 14:00:00
1007	7	107	What is NLP?	NLP helps computers understand language.	150	2026-06-07 15:20:00
1008	8	108	Explain Neural Networks	Neural networks are inspired by the human brain.	800	2026-06-08 16:10:00
1009	9	109	What is Python used for?	Python is used in web development and AI.	550	2026-06-09 17:30:00
1010	10	110	Explain Data Science	Data Science extracts insights from data.	300	2026-06-10 18:00:00
NULL	NULL	NULL	NULL	NULL	NULL	NULL

select * from Payments;

Result Grid | Filter Rows: | Edit: | Exp

payment_id	user_id	amount	payment_date	payment_status
201	1	250.00	2026-06-01 10:00:00	Completed
202	2	750.00	2026-06-02 11:00:00	Completed
203	3	450.00	2026-06-03 12:00:00	Pending
204	4	1200.00	2026-06-04 13:00:00	Completed
205	5	500.00	2026-06-05 14:00:00	Completed
206	6	850.00	2026-06-06 15:00:00	Failed
207	7	150.00	2026-06-07 16:00:00	Completed
208	8	2000.00	2026-06-08 17:00:00	Completed
209	9	650.00	2026-06-09 18:00:00	Pending
210	10	350.00	2026-06-10 19:00:00	Completed
NULL	NULL	NULL	NULL	NULL

2. Display only the username and email of all users

select username,email from Users;

Result Grid | Filter Rows:

username	email
Arun	arun@gmail.com
Bala	bala@gmail.com
Charan	charan@gmail.com
David	david@gmail.com
Ezhil	ezhil@gmail.com
Farhan	farhan@gmail.com
Gokul	gokul@gmail.com
Hari	hari@gmail.com
Imran	imran@gmail.com
John	john@gmail.com

3. Display users whose age is greater than 25

select * from Users
where age > 25;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell

user_id	username	email	age	phone	registration_date	account_status
2	Bala	bala@gmail.com	28	9876543211	2026-01-12	Active
4	David	david@gmail.com	32	9876543213	2026-02-15	Active
6	Farhan	farhan@gmail.com	30	9876543215	2026-03-10	Inactive
8	Hari	hari@gmail.com	27	9876543217	2026-04-12	Active
9	Imran	imran@gmail.com	35	9876543218	2026-05-01	Active
NULL	NULL	NULL	NULL	NULL	NULL	NULL

4. Display AI Models that are currently available

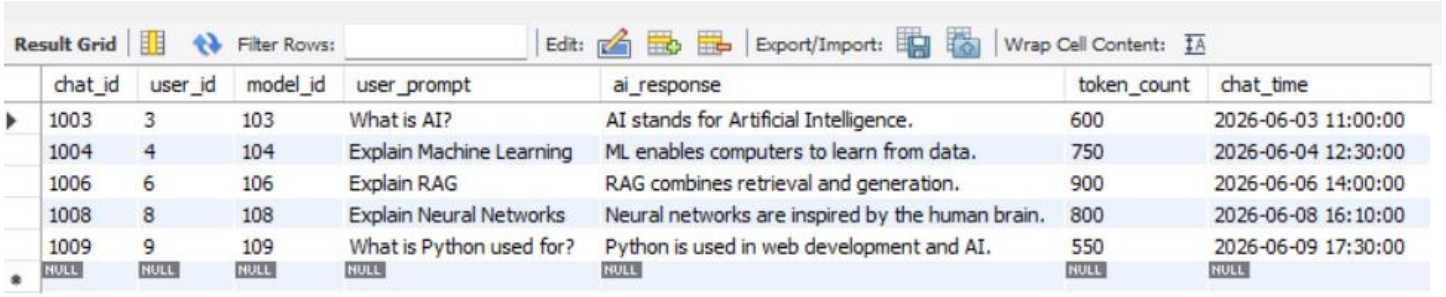
select * from AI_Models
where is_available = TRUE;

Result Grid | Filter Rows: | Edit:

model_id	model_name	model_version	is_available
101	GPT-4	4.0	1
102	GPT-3.5	3.5	1
103	Gemini	1.5	1
104	Claude	3.0	1
106	Mistral	7B	1
109	Cohere	Command-R	1
110	Phi	3.0	1
NULL	NULL	NULL	NULL

5. Display Conversations where token_count is greater than 500

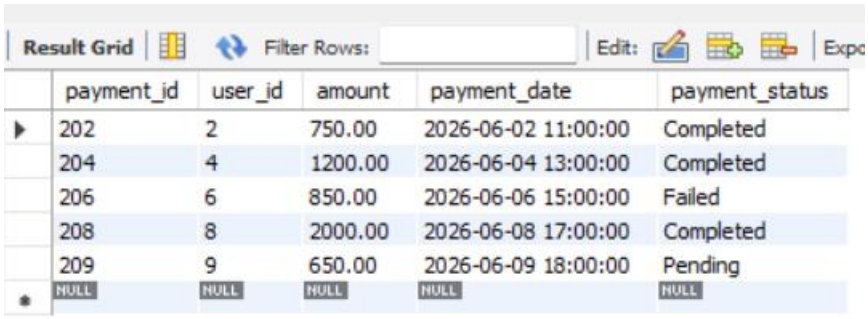
```
select * from Conversations
where token_count > 500;
```



chat_id	user_id	model_id	user_prompt	ai_response	token_count	chat_time
1003	3	103	What is AI?	AI stands for Artificial Intelligence.	600	2026-06-03 11:00:00
1004	4	104	Explain Machine Learning	ML enables computers to learn from data.	750	2026-06-04 12:30:00
1006	6	106	Explain RAG	RAG combines retrieval and generation.	900	2026-06-06 14:00:00
1008	8	108	Explain Neural Networks	Neural networks are inspired by the human brain.	800	2026-06-08 16:10:00
1009	9	109	What is Python used for?	Python is used in web development and AI.	550	2026-06-09 17:30:00
NULL	NULL	NULL	NULL	NULL	NULL	NULL

6. Display Payments with an amount greater than 500

```
select * from Payments
where amount > 500;
```



payment_id	user_id	amount	payment_date	payment_status
202	2	750.00	2026-06-02 11:00:00	Completed
204	4	1200.00	2026-06-04 13:00:00	Completed
206	6	850.00	2026-06-06 15:00:00	Failed
208	8	2000.00	2026-06-08 17:00:00	Completed
209	9	650.00	2026-06-09 18:00:00	Pending
NULL	NULL	NULL	NULL	NULL

7. Try inserting a duplicate email into the Users table. Observe the result

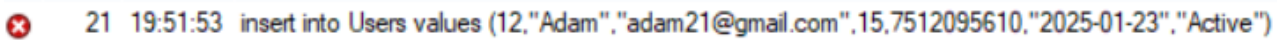
```
insert into Users values
(11,"Jeshu","david@gmail.com",23,8015012051,"2025-11-15","Active");
```



20 19:50:53 insert into Users values (11,"Jeshu","david@gmail.com",23,8015012051,"2025-11-15","Active")

8. Try inserting a user below 18 years of age. Observe the result

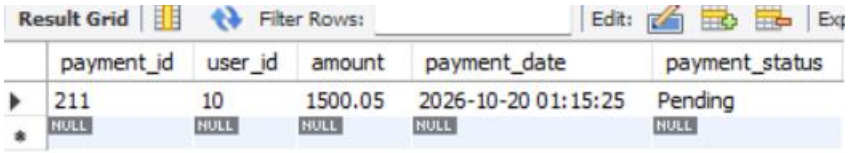
```
insert into Users values
(12,"Adam","adam21@gmail.com",15,7512095610,"2025-01-23","Active");
```



21 19:51:53 insert into Users values (12,"Adam","adam21@gmail.com",15,7512095610,"2025-01-23","Active")

9. Insert a payment without specifying payments_status.check the default value

```
insert into Payments (payment_id,user_id,amount,payment_date)
values (211,10,1500.05,"2026-10-20 01:15:25");
select * from Payments
where payment_id = 211;
```



payment_id	user_id	amount	payment_date	payment_status
211	10	1500.05	2026-10-20 01:15:25	Pending
NULL	NULL	NULL	NULL	NULL